

Toby Rodel - CV

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EDUCATION

October 2023 - March 2027	PhD (Astrophysics) at Queen's University Belfast	(In progress)
October 2022 - October 2023	MSc by Research (Physics) at University of Warwick	(Completed)
October 2019 - June 2022	BSc (Physics) at University of Warwick	(2:1)

SKILLS & EXPERTISE

Research expertise: exoplanets, transits, radial velocities, wide field photometric surveys, long-period, low mass eclipsing binaries, stellar characterisation, survey yield prediction, occurrence rates.

Instruments: *TESS*, NGTS, ULTRACAM, CORALIE, HARPS, FEROS, LCOGT

Software: TiaRA (custom pipeline), [Allesfitter](#), [PAWS](#), [iSpec](#), [isochrones](#), [Specmatch-emp](#), [AstroAriadne](#)

Programming languages: Python, SQL, HTML, CSS, $\text{\LaTeX}$, markdown

Other: public speaking and engagement, poster design, mentoring, acronym invention

RESEARCH EXPERIENCE

PhD (Queen's University Belfast with Prof Christopher Watson) 2023 - present

- Characterisation of long-period transiting planets and eclipsing binary systems including NGTS-EB-7 and TIC-65910228 / NGTS-38
- Used PAWS/iSpec and isochrones to characterise host stars and Allesfitter to fit orbital solutions to transit and radial velocity data.
- Submitted succesful ESO proposal for multiband transit observations with ULTRACAM for validation.
- Continuation of MSc work predicting survey yields, including upgrades to custom pipeline.

MSc by research (University of Warwick with Dr Daniel Bayliss) 2022 - 2023

- Developed the Transit Investigation and Recoverability Application (TiaRA) and used it to predict exoplanet discovery yields from *TESS*-SPOC data.
- Participated in eyeballing and vetting of *TESS* monotransit candidates.

BSc project (University of Warwick with Dr Matteo Brogi) 2021-2022

- Developed code to detect and fit secondary eclipses of known hot Jupiters in *TESS* data.
- Attempted albedo and heat redistribution factor determination.

PI PROGRAMMES

“Probing puffy exoplanet atmospheres with ULTRACAM” ESO, 118.25EL, ULTRACAM on NTT, 19h

SERVICE & LEADERSHIP

ESO DPR reviewer

P114 & P115

NGTS communications chair

2023 - 2025

- First chair of NGTS communications group to manage consortium promotion and outreach.
- Managed consortium social media and maintained webpage.
- Procured customised pins and stickers for consortium members.

QUB Equitea committee member

2023 - present

- Planned seminars and discussion sessions on Equity, Diversity and Inclusion topics and their relations to Astronomy

QUB ARC seminar committee member

2024-2025

- Arranged for external speakers to give seminars and hosted them.

TEACHING AND MENTORING

Co-supervisor of summer intern

June-September 2026

- Helped design cross-disciplinary project extracting SNe lightcurves from NGTS data.
- On recruitment committee to rank of 14 applicants.
- Supervisor duties focussed on working with NGTS data and liaising with consortium.

Postgraduate teaching assistant - Queen's University Belfast

2023-2024

- Problem class tutor for modules MTH1021 and PHY1002.

Mentoring of research students

2023 - Present

- Assisted and mentored QUB MSci students Sarah Jackson, Caleb Todd and Ceothàn Gormley with software, proofreading of reports and presentation practice.
- Assisted and mentored external students through the NGTS early careers group.

OUTREACH

Planetarium programmes

2022 - Present

- Setup and take-down of inflatable dome as well as presentation of shows using Stellarium at various festivals and school visits.

QUB astronomy day

2024 - present

- Volunteer 2024 & 2025, on organising committee for 2026.

Astronomy on Tap Belfast

2024 - present

- Speaker in 2024, MC and organiser in 2025.

Big bang science festival Birmingham

Summer 2023

REFEREES

Prof Christopher Watson	PhD supervisor - Queen's University Belfast
Dr Daniel Bayliss	MSc supervisor - University of Warwick
Dr Sarah Casewell	Collaborator - University of Leicester
Dr Matteo Brogi	BSc supervisor - University of Turin

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FIRST AUTHORED PUBLICATIONS

- Rodel, Toby, Daniel Bayliss, et al. (Mar. 2024). “TIARA TESS 1: estimating exoplanet yields from Years 1 and 3 SPOC light curves”. In: MNRAS 529.1, pp. 715–731. DOI: [10.1093/mnras/stae474](https://doi.org/10.1093/mnras/stae474). arXiv: [2402.07800](https://arxiv.org/abs/2402.07800) [astro-ph.EP].
- Rodel, Toby, Christopher A. Watson, et al. (Feb. 2025). “NGTS-EB-7, an eccentric, long-period, low-mass eclipsing binary”. In: MNRAS 537.1, pp. 35–55. DOI: [10.1093/mnras/stae2799](https://doi.org/10.1093/mnras/stae2799). arXiv: [2501.04523](https://arxiv.org/abs/2501.04523) [astro-ph.SR].
- Rodel, Toby, Solène Ulmer-Moll, et al. (Feb. 2026). “TIC-65910228 b / NGTS-38 b, a 180 day transiting warm super-Jupiter”. In: *arXiv e-prints*, arXiv:2602.12977, arXiv:2602.12977. DOI: [10.48550/arXiv.2602.12977](https://doi.org/10.48550/arXiv.2602.12977). arXiv: [2602.12977](https://arxiv.org/abs/2602.12977) [astro-ph.EP].

CO-AUTHORED PUBLICATIONS

- Doyle, Lauren et al. (Apr. 2024). “The TESS-SPOC FFI target sample explored with Gaia”. In: MNRAS 529.2, pp. 1802–1813. DOI: [10.1093/mnras/stae616](https://doi.org/10.1093/mnras/stae616). arXiv: [2403.02407](https://arxiv.org/abs/2403.02407) [astro-ph.SR].
- Eschen, Yoshi Nike Emilia et al. (Dec. 2024). “Viewing the PLATO LOPS2 field through the lenses of TESS”. In: MNRAS 535.2, pp. 1778–1795. DOI: [10.1093/mnras/stae2427](https://doi.org/10.1093/mnras/stae2427). arXiv: [2409.13039](https://arxiv.org/abs/2409.13039) [astro-ph.EP].
- Hawthorn, Faith et al. (Feb. 2024). “TESS duotransit candidates from the Southern Ecliptic Hemisphere”. In: MNRAS 528.2, pp. 1841–1862. DOI: [10.1093/mnras/stad3783](https://doi.org/10.1093/mnras/stad3783). arXiv: [2310.17268](https://arxiv.org/abs/2310.17268) [astro-ph.EP].
- O’Brien, Sean M. et al. (Dec. 2025). “NGTS-EB-8: A Double-lined Eclipsing M+M Binary Discovered by Citizen Scientists”. In: AJ 170.6, 316, p. 316. DOI: [10.3847/1538-3881/ae0e0d](https://doi.org/10.3847/1538-3881/ae0e0d). arXiv: [2510.11575](https://arxiv.org/abs/2510.11575) [astro-ph.SR].
- Ulmer-Moll, S. et al. (Nov. 2025). “Detection and characterisation of a 106-day transiting Jupiter: TOI-2449 b/NGTS-36 b”. In: A&A 703, A258, A258. DOI: [10.1051/0004-6361/202555168](https://doi.org/10.1051/0004-6361/202555168). arXiv: [2509.15424](https://arxiv.org/abs/2509.15424) [astro-ph.EP].
- Apergis, Ioannis et al. (Jan. 2026). “High-precision photometry with a scientific CMOS camera: II on-sky testing of the Marana camera at the NGTS facility”. In: *RAS Techniques and Instruments* 5, rzag022, rzag022. DOI: [10.1093/rasti/rzag022](https://doi.org/10.1093/rasti/rzag022). arXiv: [2603.16361](https://arxiv.org/abs/2603.16361) [astro-ph.IM].
- Kendall, Alicia et al. (Apr. 2026). “A 43 d transiting Neptune and two 25 d Saturns from TESS, NGTS, and ASTEP”. In: MNRAS 547.2, staf2189, staf2189. DOI: [10.1093/mnras/staf2189](https://doi.org/10.1093/mnras/staf2189). arXiv: [2512.07716](https://arxiv.org/abs/2512.07716) [astro-ph.EP].

CONFERENCE PRESENTATIONS

- “Predicting the yield of long-period Planets in TESS”, UK Exoplanet Meeting, University College London, August 2023, Poster and pop-up talk
- “Putting a TIARA on SPOC: Predicting long-period planet yields from TESS”, UK Exoplanet Meeting, University of Birmingham, April 2024, Poster and pop-up talk
- “Putting a TIARA on SPOC”, TESS Science Conference 3, Massachusetts Institute of Technology, August 2024, Contributed talk and poster
- “Long period planets from TESS and NGTS”, Irish National Astronomy Meeting, University of Galway, August 2024, Contributed talk
- “NGTS-EB-7: One of the longest period EBLM systems ever found”, UK Exoplanet Meeting, University of Leeds, April 2025, Poster and pop-up talk
- “NGTS-EB-7: One of the longest period EBLM systems ever found”, Binary stars in the space era, University of Keele, July 2025, Poster and pop-up talk
- “How good is TESS at detecting long-period exoplanets”, From Transits to Trends: the Next Decade of Long-Period Exoplanets, University of New Mexico, August 2025, Contributed talk
- “Finding hidden long-period planets with TESS”, UK Exoplanet Meeting, University of Bristol, April 2026, Contributed talk